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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The pancreas is an organ with many functions.

A student observed some cells of the pancreas using an electron microscope.

The student described one organelle that was seen as

> "several curved membrane-bound sacs of decreasing size".

(a) (i) Which organelle was seen?

(1)

☒ **A** centriole

☒ **B** Golgi apparatus

☒ **C** rough endoplasmic reticulum

☒ **D** smooth endoplasmic reticulum

(ii) Explain why this organelle cannot be seen using a light microscope.

(2)

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(iii) Give **two** differences between an organ and a tissue.

(2)

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(b) Plant cells can also be studied using an electron microscope.

Name **two** structures present in animal cells that are not present in a plant cell.

(2)

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1 .....

2 .....

(c) Cells in the pancreas use amino acids to synthesise polypeptides.

These cells also secrete enzymes and glycoproteins into the small intestine.

Cells from the pancreas, containing amino acids labelled with a fluorescent marker, were used in an investigation. The fluorescent marker looked green when seen with a special microscope.

The percentage of green fluorescence inside and outside the cells was measured at the start and after 60 minutes.

The results of this investigation are shown in the table.

Time / min	Percentage of green fluorescence (%)	
	Inside the cells	Outside the cells
0	100	0
60	38	62

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Explain the results of this investigation.

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(Total for question = 11 marks)

Q2.

Meiosis increases genetic variation through independent assortment and crossing over.

(a) (i) In which stage of meiosis does independent assortment occur?

(1)

☒ **A** metaphase I

☒ **B** metaphase II

☒ **C** telophase I

☒ **D** telophase II

(ii) In which stage of meiosis does crossing over occur?

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(1)

☒ **A** metaphase I

☒ **B** metaphase II

☒ **C** prophase I

☒ **D** prophase II

(b) Genetic variation increases due to the production of new combinations of alleles.

A cell has gene A and gene B with alleles A, a and B, b.

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Meiosis could produce the following percentages of gametes containing the combinations of alleles shown in the table.

Combination of alleles	Percentage of total gametes produced (%)
AB	48
Ab	2
aB	2
ab	48

Explain why the percentages are not 25% for each combination of alleles.

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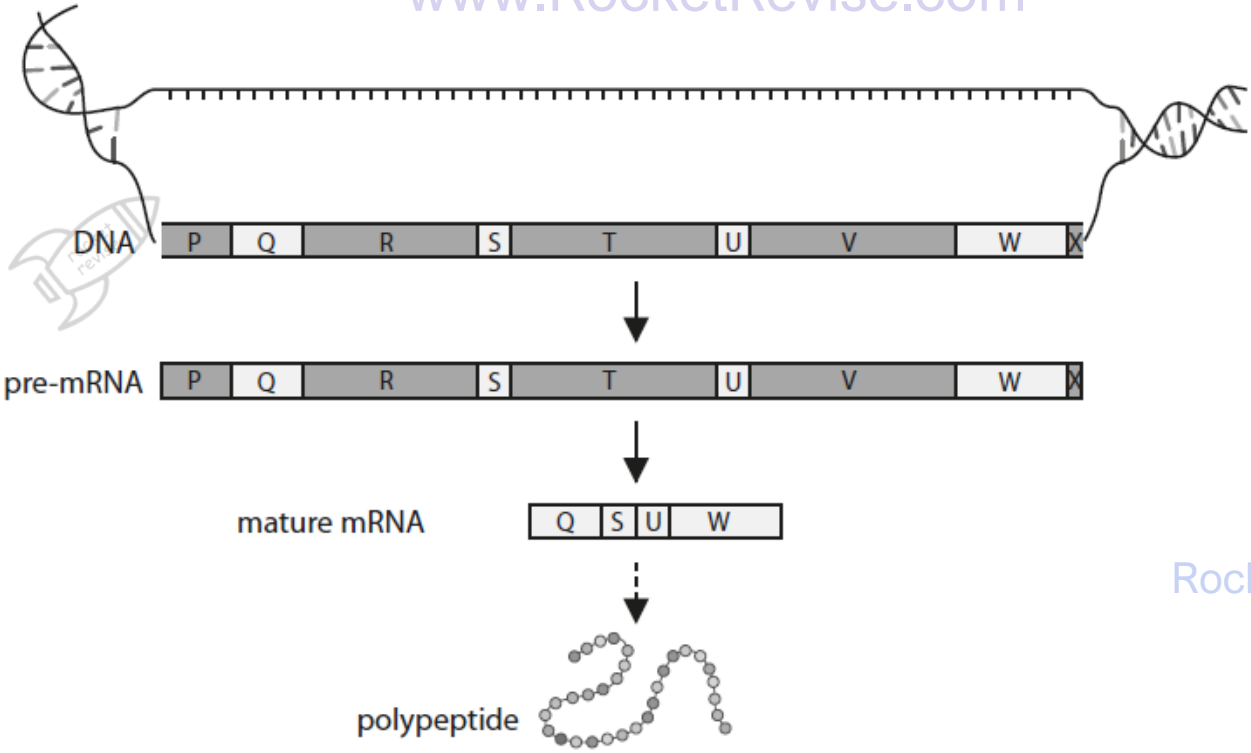
(c) More than one type of protein can be synthesised from the RNA produced from one gene.

The diagram shows parts of this process.

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Describe how more than one type of protein can be synthesised from the RNA produced from one gene.

Use the information in the diagram to support your answer.

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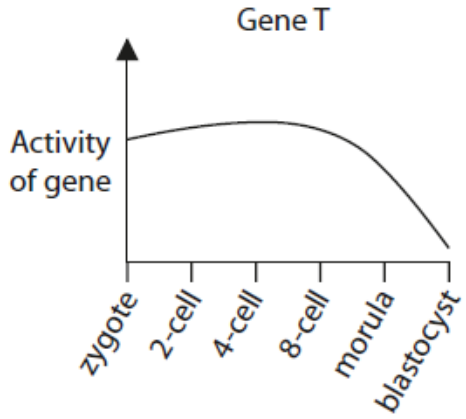
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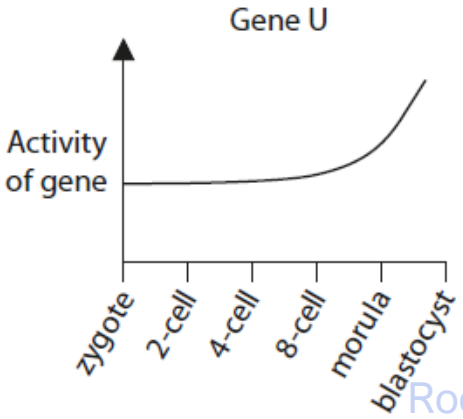
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(d) The graphs show the activity of two genes, T and U, as a mouse zygote (fertilised egg cell) develops into a blastocyst.





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Comment on the changes in the activity of these two genes.

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(Total for question = 13 marks)

Q3.

Some of the characteristics of an organism are due to polygenic inheritance.

(a) The photograph shows some Chinese white kidney beans.

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The mass of individual Chinese white kidney beans varies. This is partly due to polygenic inheritance.

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(i) State what is meant by the term **polygenic inheritance**.

(2)

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(ii) Which of the following graphs represents the distribution of mass of these beans?

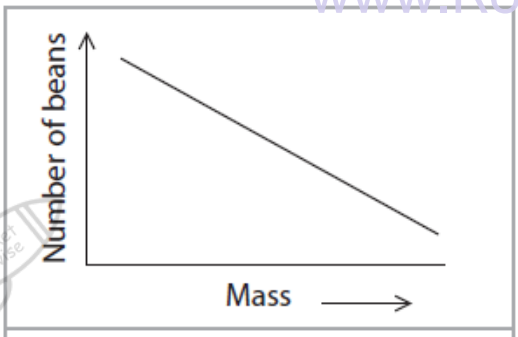
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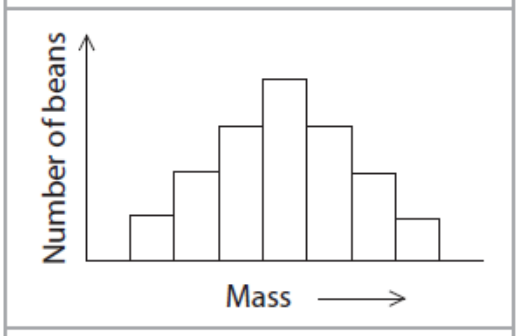
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☒ A

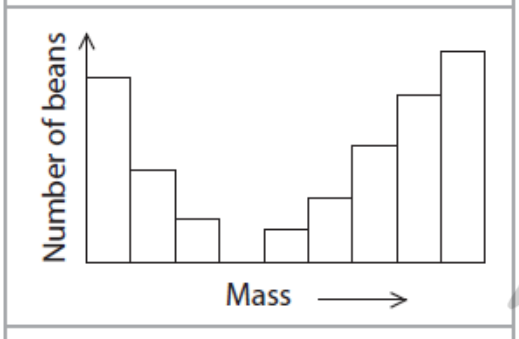


☒ B

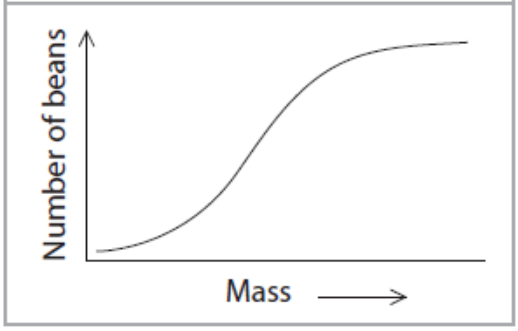


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☒ C



☒ D



(b) The inheritance of height is another example of polygenic inheritance.

It is possible to predict how tall children could become when they reach adulthood.

The prediction uses the following steps.

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1. Add the mother's height (cm) to the father's height (cm).
2. Add 13 cm if the child is a boy or subtract 13 cm if the child is a girl.
3. Divide by two.

(i) A woman with a height of 1.59 m has a daughter. The father is 1.76 m tall.

Calculate the predicted adult height of their daughter.

(1)

Answer ..... m

(ii) Explain why their daughter may not grow to be this height.

(3)

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(Total for question = 7 marks)

Q4.

Insulin is a protein produced in beta cells in the pancreas.

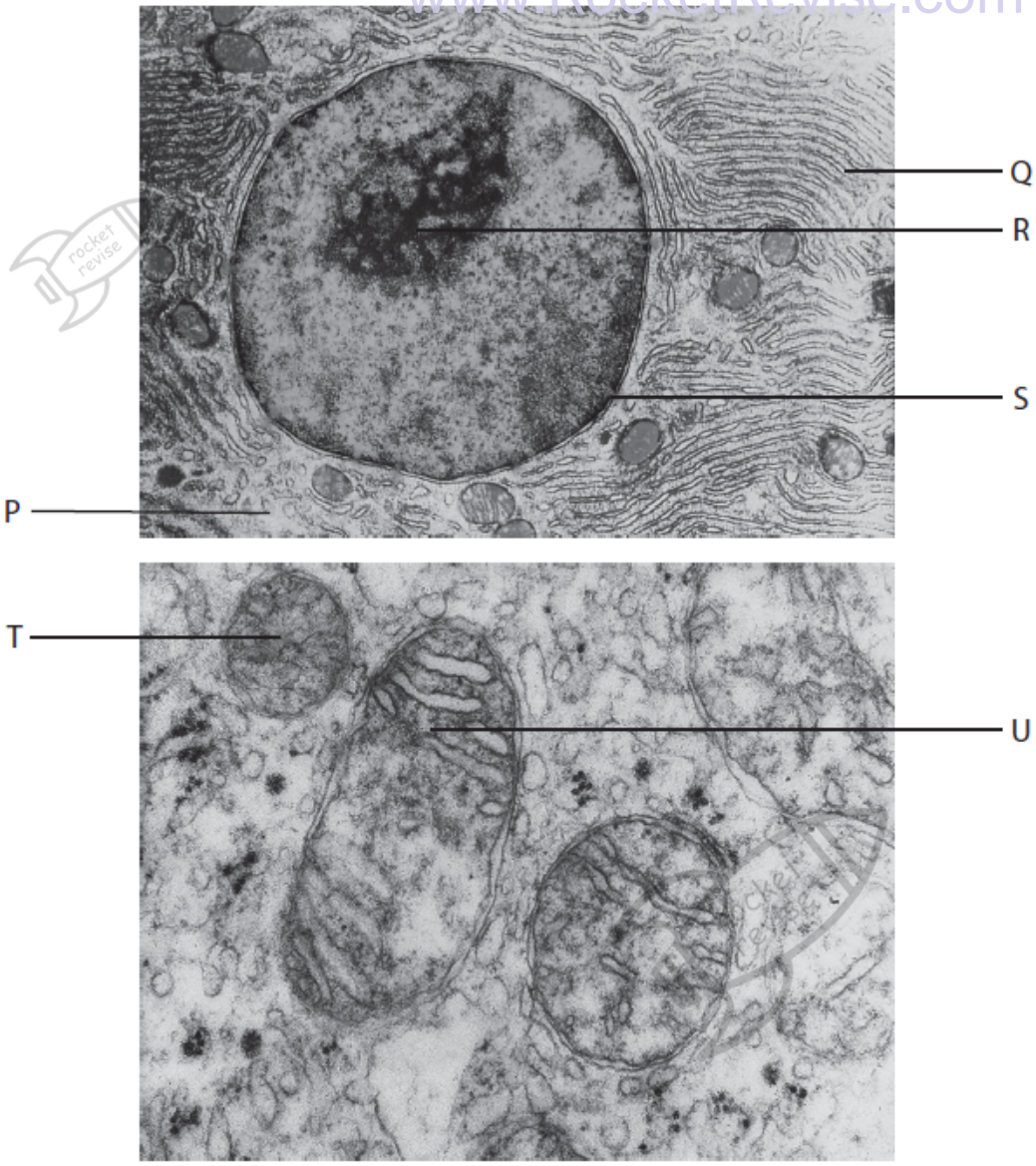
Ribosomes are involved in the production of insulin.

(a) The photographs show part of a beta cell from the pancreas, as seen using an electron microscope.

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(Source: © Don Fawcett/Science Photo Library
© Martin M. Rotker/Science Photo Library)

(i) Which labelled structure produces ribosomes?

(1)

☒ **A** P

☒ **B** Q

☒ **C** R

☒ **D** S

(ii) The structures labelled T and U are the same type of organelle.

Suggest why these structures look different in this photograph.

(2)

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(iii) A student measured the length of the structure labelled U as 52 mm.

The actual length of this structure is 3.0 µm.

Calculate the magnification of this electron microscope image.

(2)

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Answer .....

(b) Describe how the protein insulin is transported from the ribosomes and then secreted by a beta cell.

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(c) Explain why beta cells can produce insulin but other cells in the pancreas do not.

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Q5.

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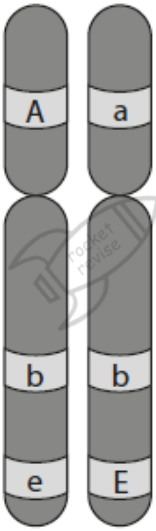
Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The nucleus of a cell contains chromosomes.

The diagram shows three genes present on a pair of chromosomes.

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(a) (i) How many of the following statements about this diagram are correct?

- there are three gene loci
- the pair of chromosomes will be separated into different cells following mitosis
- one chromosome was inherited from the mother and one chromosome was inherited from the father

(1)

☒ **A** none

☒ **B** one

☒ **C** two

☒ **D** three

(ii) Describe what will happen to these chromosomes when they enter the interphase stage of the cell cycle.

(2)

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(b) Crossing over between pairs of chromosomes increases genetic variation.

(i) Name the stage of meiosis in which crossing over begins.

(1)

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(ii) How many of the following combinations could result from one crossover of the chromosomes shown in the diagram?

- AbE
- ABe
- abE
- abe

(1)

☒ **A** one

☒ **B** two

☒ **C** three

☒ **D** four

(iii) Explain why alleles b and e are more likely to be inherited together than alleles A and e.

(2)

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(Total for question = 7 marks)

Q6.

Egg cells are produced by many organisms.
The development of fish embryos can be studied using fertilised egg cells.
The photograph shows part of a beluga fish from the Caspian Sea.

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(Source: © Design Pics / Alamy Stock Photo)

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(a) Explain how an egg cell is specialised for its function.

(2)

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(b) A student studied beluga fish cells undergoing mitosis and meiosis.

(i) Compare and contrast metaphase in mitosis and meiosis.

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(ii) Some of the cells undergoing mitosis were from a beluga morula whereas some of the cells

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were from a beluga blastocyst.

Explain how the cells of the beluga morula change as they develop into the cells of the blastocyst.

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(c) Chickens produce eggs.
The effect of the age of a chicken on the mass of the eggs it laid was investigated.

A statistical test can be used to see if there is a significant correlation between the age of a chicken and the mass of her eggs.

The value generated by this statistical test will always be between −1.0 to +1.0.

The table shows the mass of eggs collected from chickens of different ages.

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Age / weeks	Rank of age	Mass of egg / g	Rank of mass of egg	D	D^2
36	1	2.27	8	−7	49
39	2	1.91			
42	3	2.18	5.5	−2.5	6.25
45	4	2.28	9	−5	25
48	5	2.12	4	1	1
51	6	2.19	7	−1	1
54	7	2.18	5.5	1.5	2.25
57	8	2.09	3	5	25
60	9	2.03	2		

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(i) Calculate the value for $\sum D^2$.
Use the table to help you.

(2)

$\sum D^2 =$ .....

(ii) Calculate the correlation coefficient, r_s value, for this data using the formula:

(2)

$$r_s = 1 - \frac{6\left(\sum D^2\right)}{n\left(n^2 - 1\right)}$$

Answer .....

(iii) The null hypothesis is that there was no significant correlation between the age of the chicken and the mass of the eggs laid.

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Another scientist replicated this study and gained a correlation coefficient value of 0.38.
The table shows the critical values at the 95% probability level.

Number of pairs of data	Critical values at the 0.05 % probability level
8	0.738
9	0.683
10	0.648

Explain whether the null hypothesis should be accepted or rejected.

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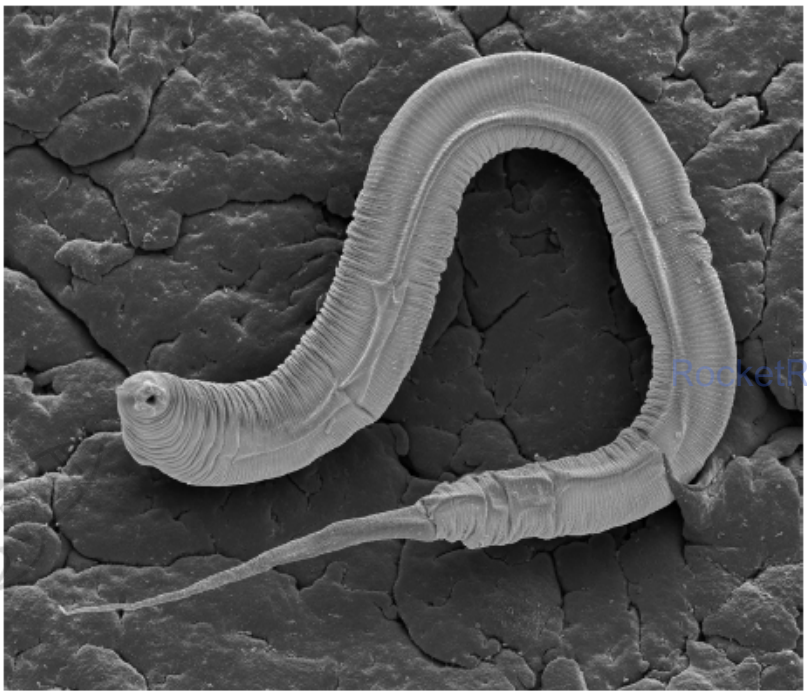
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(Total for question = 15 marks)

Q7.

Variation between organisms occurs as a result of sexual reproduction and epigenetic modification.

The photograph shows a nematode, as seen using an electron microscope.



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(Source: © Leonid Serebrennikov / Alamy Stock Photo)

* (a) Animals of this species are either male or hermaphrodites.

Nematodes that inherit one sex chromosome are male.

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Nematodes that inherit two sex chromosomes are hermaphrodites.

Hermaphrodites can produce both egg cells and sperm. Approximately 300 sperm are produced first and stored inside the nematode. After this, egg cells are produced.

The sperm of a hermaphrodite nematode can fertilise its own egg cells. This is called self-fertilisation.

However, when a hermaphrodite nematode mates with a male nematode, the sperm of the male fertilises the egg cells. This results in 700 more egg cells being fertilised than when a hermaphrodite self-fertilises. Half of these fertilised egg cells will result in male offspring.

Explain how the reproductive behaviour of these nematodes increases their chance of survival.

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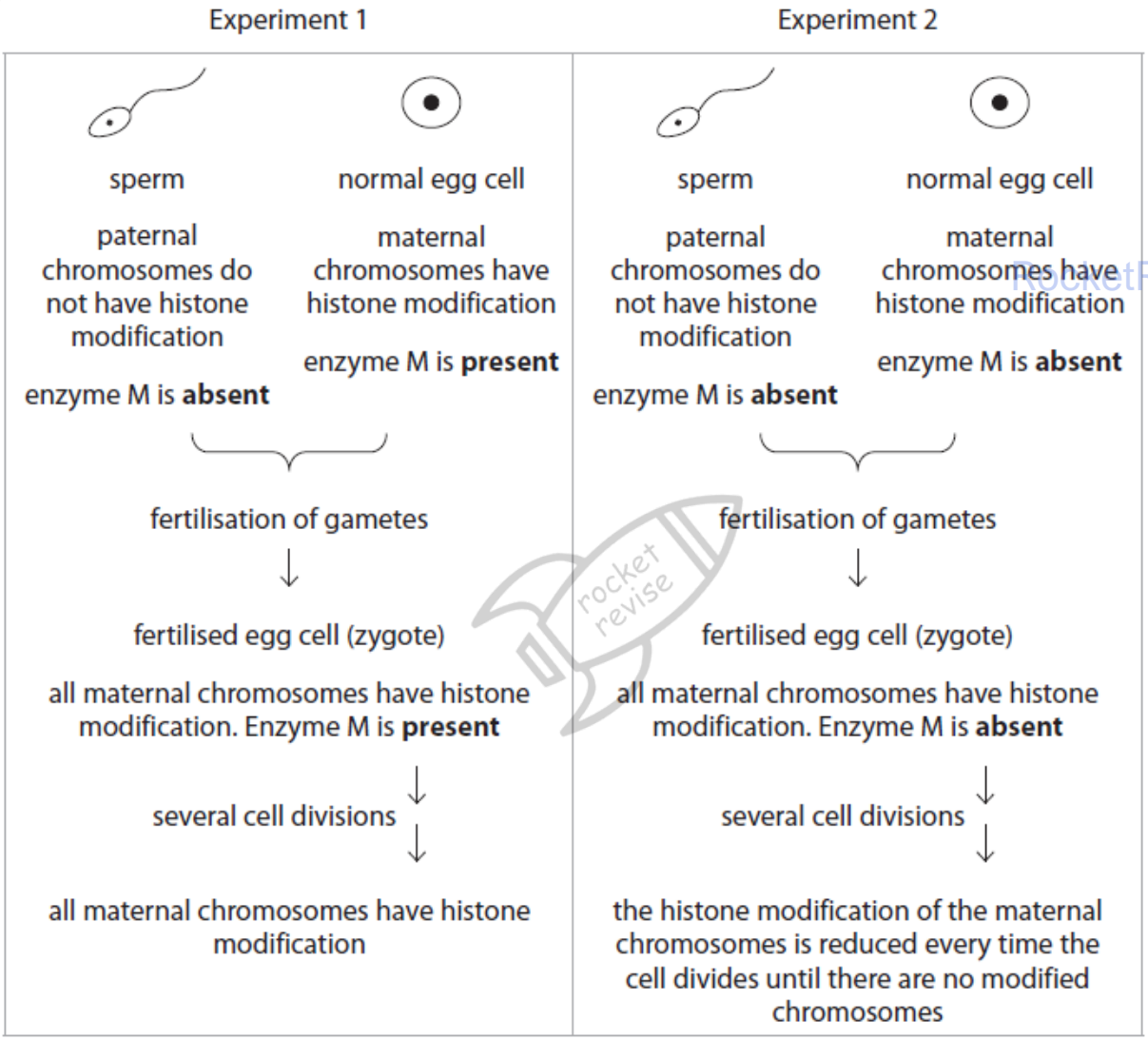
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(b) In nematodes, histone modification can be passed onto embryos.
Enzyme M causes this modification.
The diagram shows two experiments that demonstrated this.



Deduce how histone modification can be passed onto embryos.

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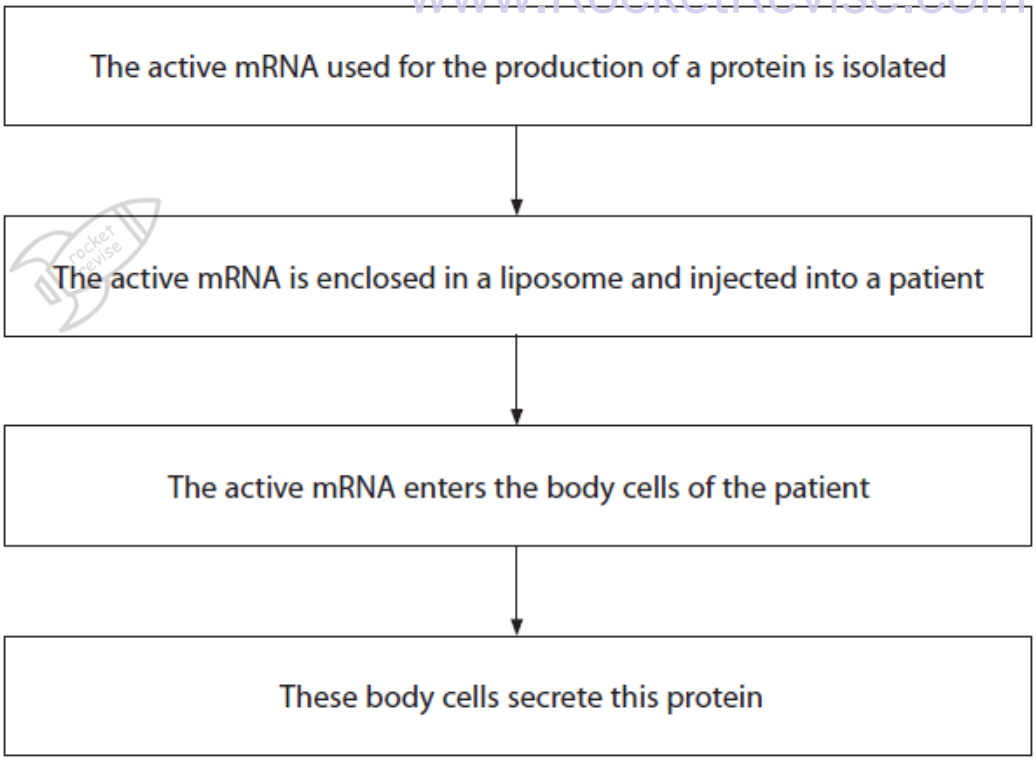
Q8.

A new vaccine has been developed that contains active mRNA.

The flow diagram shows how this vaccine causes body cells to produce a new protein.

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(a) The diameter of one spherical liposome was 75 nm.

Calculate the volume of this liposome using the formula:

$$V = \frac{4}{3}\pi r^3$$

Give your answer in standard form.

(2)

Answer ..... nm^3

(b) Active mRNA is formed after post-transcriptional modification of pre-mRNA.

The structure of pre-mRNA produced from transcription is different from the structure of active mRNA.

Compare and contrast the structures of pre-mRNA and active mRNA.

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Q9.

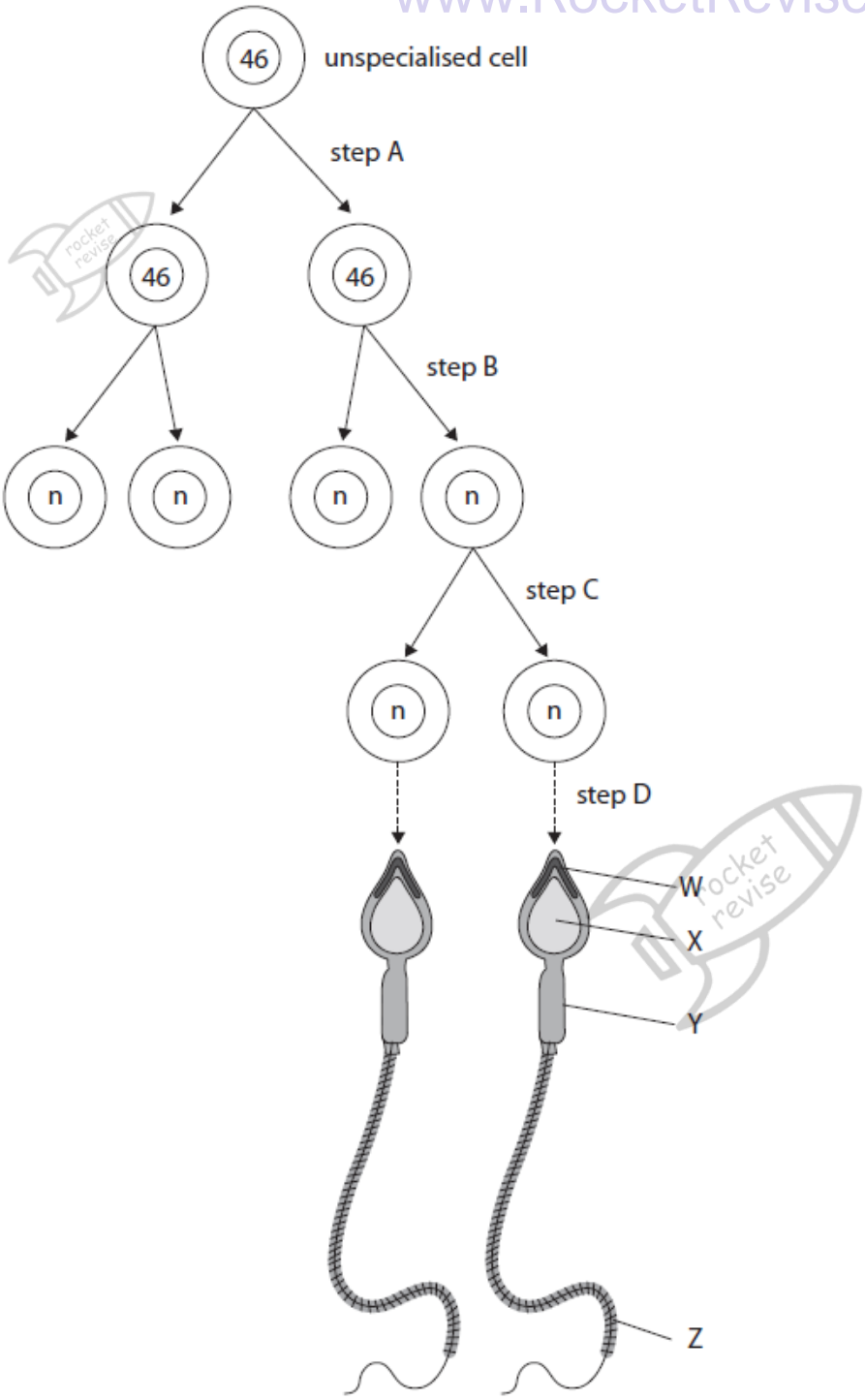
Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows some of the steps involved in the production of human sperm cells from an unspecialised cell. This process involves both mitosis and meiosis.

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(a) (i) The unspecialised cell is diploid.

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State the types of division occurring at step A and step B.

(2)

step A .....

step B .....

(ii) Which part of the sperm cell contains enzymes that are released to break down the zona pellucida?

(1)

☐ **A** W

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☒ **B** X

☒ **C** Y

☒ **D** Z

(b) A sperm cell is a specialised cell.

Explain what happens at step D for the cell to become a specialised sperm cell.

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(c) The nucleus of a sperm cell fuses with the nucleus of an egg cell to form a zygote. The zygote develops into a blastocyst.

(i) Draw a diagram to show the arrangement of the cells of a blastocyst.

(2)

(ii) The blastocyst contains some pluripotent stem cells.

Which of the following describes these pluripotent stem cells?

(1)

☒ **A** adult stem cells that can give rise to most types of cell in the body, including totipotent

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cells

☒ **B** adult stem cells that can give rise to most types of cell in the body, excluding totipotent cells

☒ **C** embryonic stem cells that can give rise to most types of cell in the body, including totipotent cells

☒ **D** embryonic stem cells that can give rise to most types of cell in the body, excluding totipotent cells

(d) The photograph shows a Mexican salamander.



(Source: © Imagebroker/Alamy Stock Photo)

Salamanders can regenerate damaged limbs and organs throughout their lives.

An ethics committee has approved the use of stem cells taken from salamanders.

Suggest why the use of stem cells taken from salamander embryos was approved.

(2)

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(Total for question = 13 marks)

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Q10.

Mitosis and the cell cycle are involved in producing new cells.

(a) Describe the role of the cell cycle in producing new cells.

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(b) Uncontrolled cell division can result in the formation of a tumour.

The RB gene is a tumour suppressor gene.

The RB protein produced from this gene inhibits tumour growth.

Lung cancer can occur when this gene has increased methylation compared with the RB gene in normal lung cells.

Explain why increased methylation of this gene could cause the growth of a tumour.

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(c) The cell cycle can be affected by various chemicals.

One drug, used to treat various types of cancer, prevents the shortening of spindle fibres.

Explain how preventing the shortening of spindle fibres affects mitosis.

(2)

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* (d) Paclitaxel is a drug used to treat some tumours.

In one study, the effect of paclitaxel on the cell cycle in human lung tumours was investigated.

The table shows the results of this study.

Concentration of paclitaxel / nmol dm^{-3}	Duration of mitosis / hours	Mitotic index of cancerous cells in lung tissue (%)
0	1	2
10	4	17
100	9	38
1 000	21	48
10 000	24	59

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In another study, the effect of paclitaxel on the growth of breast tumours in mice was investigated. The effect of paclitaxel was compared with drug X and with a placebo.

For each group the increase in mass of the tumour was measured 21 days after starting treatment.

Three groups of mice with breast tumours were given the following treatments:

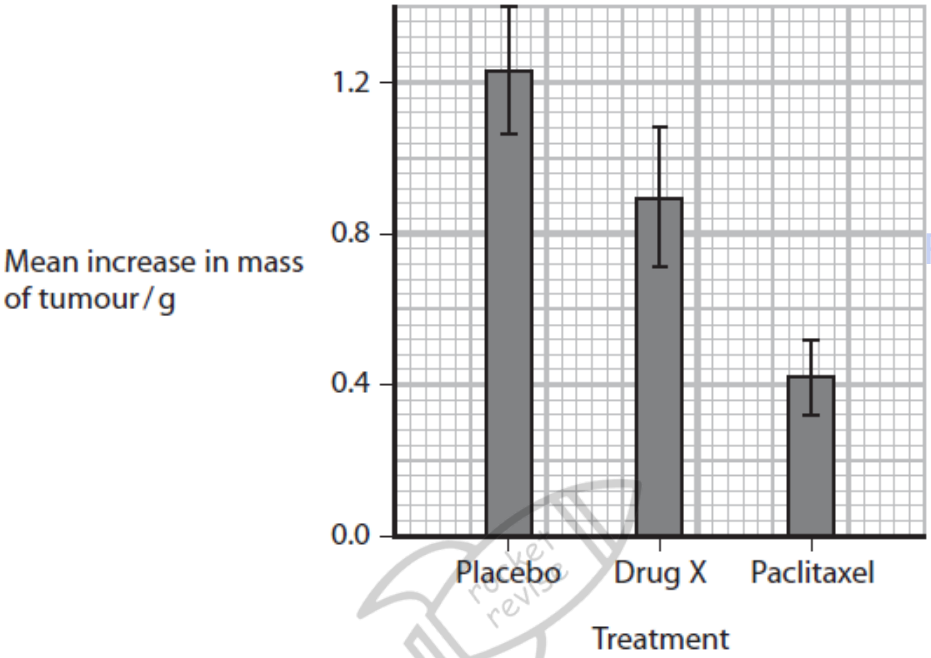
- Group 1 received a placebo

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- Group 2 received drug X
- Group 3 received paclitaxel.

The graph shows the results from this study.



Evaluate the effectiveness of paclitaxel.

Use the information given to support your answer.

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Q11.

The phenotype of an organism is affected by a number of factors.

(a) State what is meant by the term **phenotype**.

(1)

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(b) Coat colour in rabbits is determined by multiple alleles.

The table gives some information about coat colour in rabbits.

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Type of rabbit	Coat colour of rabbit	.....
Black 	black all over	CC
Chinchilla 	grey all over	$c^{ch}c^{ch}$
Himalayan 	white body black ears, face, feet and tail	$c^{h}c^{h}$
Albino 	white all over	cc

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(i) Complete this table by writing a suitable heading for the right-hand column.

(1)

(ii) Which row of the table gives the correct number of genes and alleles for coat colour in these rabbits?

(1)

	Number of genes for coat colour	Number of alleles for coat colour
☒ **A**	1	1
☒ **B**	1	4
☒ **C**	4	1
☒ **D**	4	4

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(c) Height is one phenotype of an elephant.

The photograph shows an African elephant.

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Source: Caroline Wilcox

Male African elephants range in height from 3.2 m to 4.0 m.

Female African elephants range in height from 2.2 m to 2.6 m.

(i) Which row of the table names the types of graph that should be drawn to show sex and height variation in a population of African elephants?

(1)

	Sex	Height
☒ **A**	bar chart	bar chart
☒ **B**	bar chart	histogram
☒ **C**	histogram	bar chart
☒ **D**	histogram	histogram

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(ii) Calculate how many times bigger the male African elephant is than the female African elephant.

(2)

Answer .....

(Total for question = 6 marks)

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Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some of the characteristics of an organism are due to polygenic inheritance.

(a) Every person has a Rhesus characteristic.

People who are Rhesus positive have one or more of three types of Rhesus antigen C, D or E, on the cell surface membrane of their red blood cells.

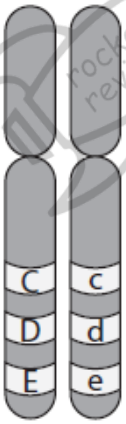
Each Rhesus antigen is coded by a dominant allele of gene C or gene D or gene E.

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People who are Rhesus negative for one of these genes will have two recessive alleles.

For example, a person who is Rhesus D negative will have inherited two recessive d alleles.

The diagram shows the loci of these three genes on a pair of chromosomes.



(i) State what is meant by the term **allele**.

(1)

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(ii) Which of the following is the genotype of a person who is Rhesus negative for all three genes?

(1)

☒ **A** ccddee

☒ **B** ccddEe

☒ **C** CcDdEe

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☒ **D** CCDDEE

(iii) Rhesus antigens are glycoproteins.

Describe how a Rhesus antigen would be produced from its polypeptide chain and transported to the cell surface membrane.

(5)

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(b) Seed colour in wheat is an example of polygenic inheritance. There are three genes thought to affect
seed colour (A, B, C). Each gene has a dominant and recessive allele.

A cross between two heterozygous plants (AaBbCc) was carried out.

The seed colour of the offspring was recorded.

The table below shows the phenotype of the wheat seed colour, the number of recessive alleles and the frequency of seed colour in the offspring.

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Colour of wheat seed	Number of recessive alleles	Frequency of seed colour
White	6	1
Very light red	5	6
Light red	4	15
Intermediate red	3	20
Red	2	15
Moderate red	1	6
Dark red	0	1

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Explain this variation in the colour of wheat seeds.

(3)

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(Total for question = 10 marks)

Q13.

Skin colour is an example of continuous variation in some species.

In one species skin colour is polygenic and controlled by three genes: A, B, and C.

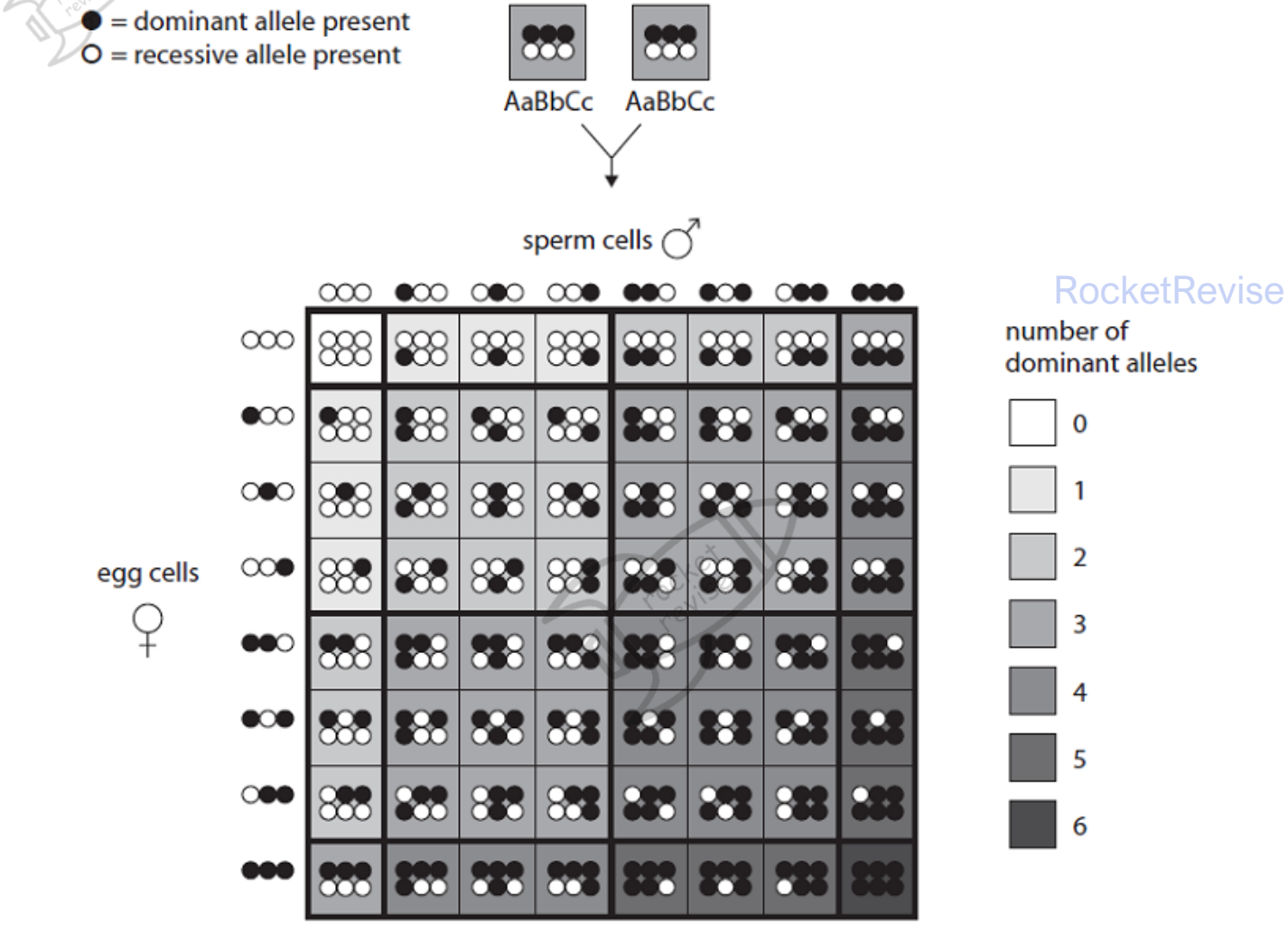
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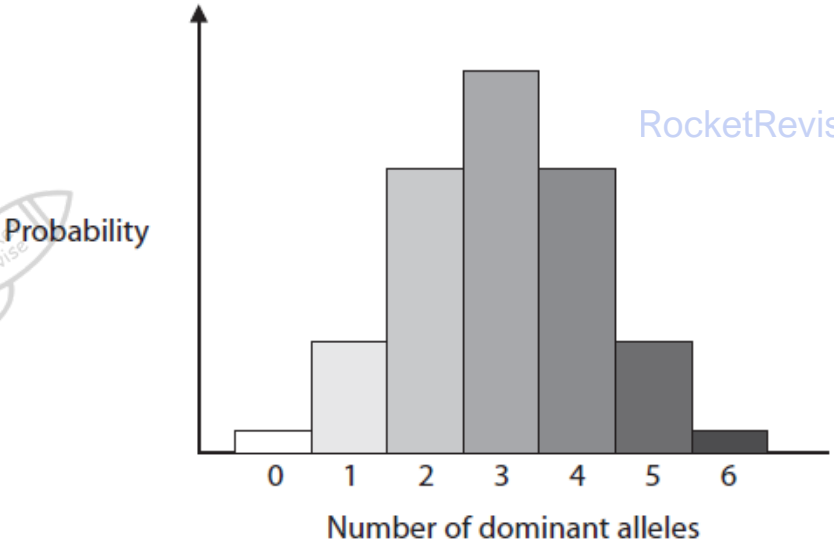
Two individuals heterozygous for each of these three genes were crossed.

The diagram shows the possible allele combinations present in the gametes and the offspring produced in this cross.

The shading indicates the darkness of the skin colour of the offspring.



The graph shows the probability of the skin colours of the offspring from this cross.



*(a) Explain why there is a large variation in the skin colour of the offspring produced from this cross.

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Use information from the question, and your own knowledge, to help support your answer.

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(b) Suggest **two** reasons why individuals that have inherited the genotype AaBbCc, may have a darker skin colour than other individuals with the same genotype.

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(Total for question = 8 marks)

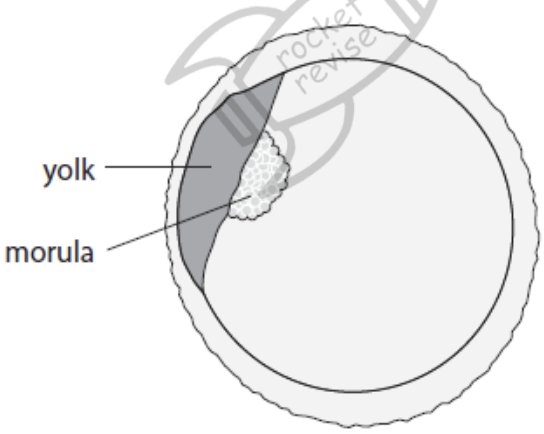
Q14.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Cells can divide by mitosis.

The diagram shows a morula beginning to develop on the yolk of a fertilised fish egg cell.



(a) (i) Which of the following describes a morula?

(1)

☒ **A** a hollow ball of pluripotent stem cells

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☒ **B** a hollow ball of totipotent stem cells

☒ **C** a solid ball of pluripotent cells

☒ **D** a solid ball of totipotent cells

(ii) Suggest **one** function of the yolk in the fertilised fish egg cell.

(1)

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(b) As the fish embryo develops, different types of body cells and tissues are formed. One type of tissue that develops is skeletal muscle tissue.

All the cells in the skeletal muscle tissue contain the same genes as the morula cells.

Explain how these skeletal muscle cells can contain the same genes as the morula cells but be different in structure and function.

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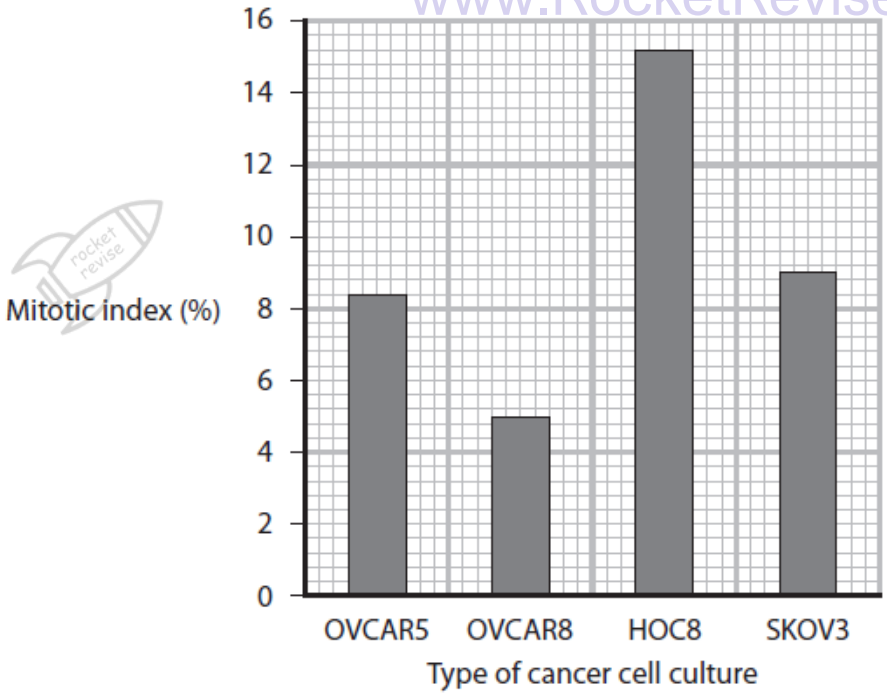
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(c) The mitotic index of four types of cancer cell culture was investigated.

The graph shows the results of this investigation.

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(i) Two thousand OVCAR8 cells were studied.

Calculate how many OVCAR8 cells were in interphase.

(2)

Answer .....

(ii) Which cancer cell culture has the highest proportion of cells undergoing nuclear division?

(1)

☒ **A** HOC8

☒ **B** OVCAR5

☒ **C** OVCAR8

☒ **D** SKOV3

(iii) Explain why some people think that research into cancer treatments should not use embryonic stem cells.

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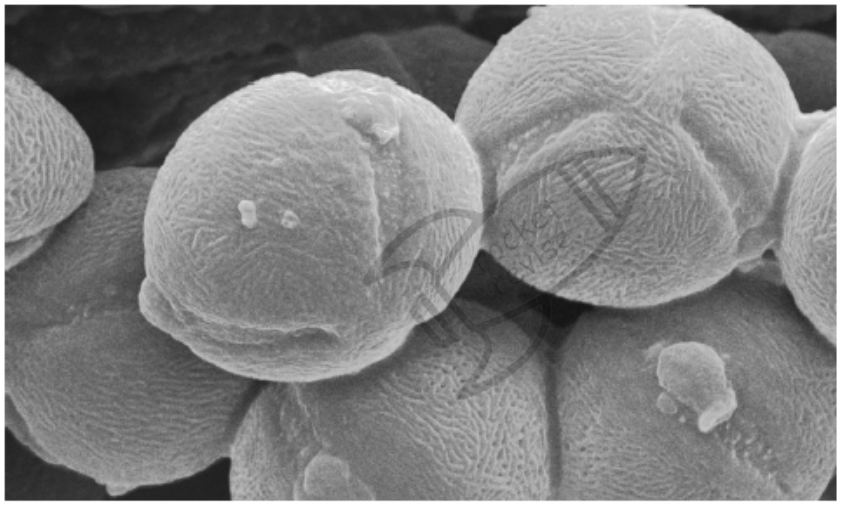
(Total for question = 12 marks)

Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The photograph shows some pollen grains, as seen using an electron microscope.



(Source: BSIP SA / Alamy Stock Photo)

Scale bar ⊢——⊣
20 µm

(a) Calculate the magnification of this photograph.

Give your answer to **two** significant figures.

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(2)

Answer .....

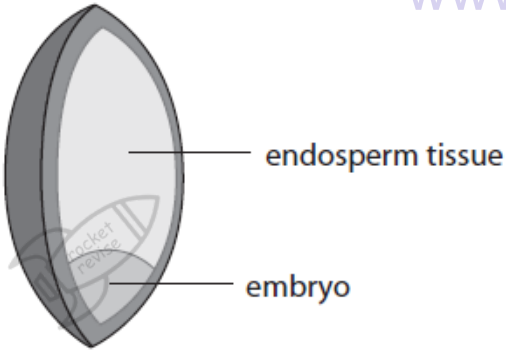
(b) A pollen grain can produce a pollen tube that contains a generative nucleus.

Plant fertilisation results in the production of an embryo and endosperm tissue.

The diagram shows the location of an embryo and endosperm tissue in a seed.

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Describe how the generative nucleus results in the production of an embryo and endosperm tissue in a seed.

(3)

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(c) Flower colour and pollen grain size are genetically determined in foxglove plants.

The linkage of the genes for flower colour and pollen grain size has been investigated.

(i) State what is meant by the term **linkage** with reference to these two genes.

(1)

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(ii) A statistical test can be used to analyse observed phenotype frequencies and expected phenotype frequencies for the foxglove plants.

Which of the following statistical tests would be used?

(1)

☒ **A** Chi squared

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☒ **B** correlation coefficient

☒ **C** standard deviation

☒ **D** Student's t-test

(Total for question = 7 marks)

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Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Eye colour in humans is an example of polygenic inheritance.

Two of the genes giving rise to this characteristic are found on chromosome 15.

(a) There is more than one version of each of the genes involved.

Which term is used for different versions of a gene?

(1)

☒ **A** allele

☒ **B** locus

☒ **C** phenotype

☒ **D** polygenic

(b) Inheritance of different versions of these two genes shows genetic linkage.

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Explain how the phenotypes of an individual can show genetic linkage in the inheritance of two genes.

(2)

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(c) One gene can give rise to more than one type of protein through post-transcriptional changes to the mRNA.

One of these changes is the removal of introns.

Explain how different proteins can be synthesised from this mRNA that has had introns removed.

(3)

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(Total for question = 6 marks)

Q17.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A pike is a fish found in freshwater rivers.

The photograph shows a male and a female pike.

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(Source: © Ross Jolliffe/Alamy Stock Photo)

Female pike lay egg cells on the leaves of river plants.
Male pike then release sperm cells that will fertilise the egg cells.

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(a) (i) An unfertilised egg cell is 2.3 mm in diameter.

Calculate the volume of the egg cell, using the formula:

$$V = \frac{4}{3}\pi r^3$$

Give your answer to **one** decimal place.

(1)

Answer ..... mm^3

(ii) An unfertilised pike egg cell has a larger volume than a pike sperm cell.

Explain why an egg cell has a larger volume than a sperm cell.

(2)

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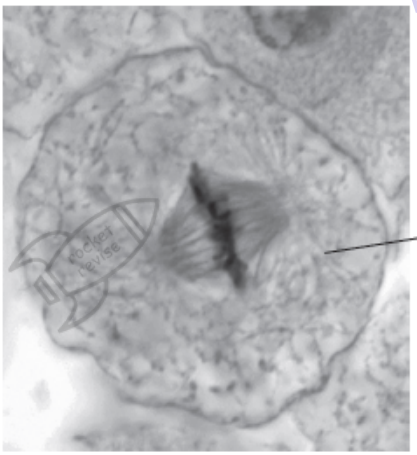
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(b) The photograph shows a cell from the blastocyst stage of development.

The blastocyst cell was treated to show the stage of mitosis.

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(Source: © HERVE CONGE, ISM/SCIENCE PHOTO LIBRARY)

(i) Which stage of mitosis is shown in this blastocyst cell?

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(1)

☒ **A** cytokinesis

☒ **B** metaphase

☒ **C** prophase

☒ **D** telophase

(ii) State how the blastocyst cell was treated to show the stage of mitosis.

(1)

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(iii) A blastocyst develops from a morula.

Compare and contrast the structure of a morula with the structure of a blastocyst.

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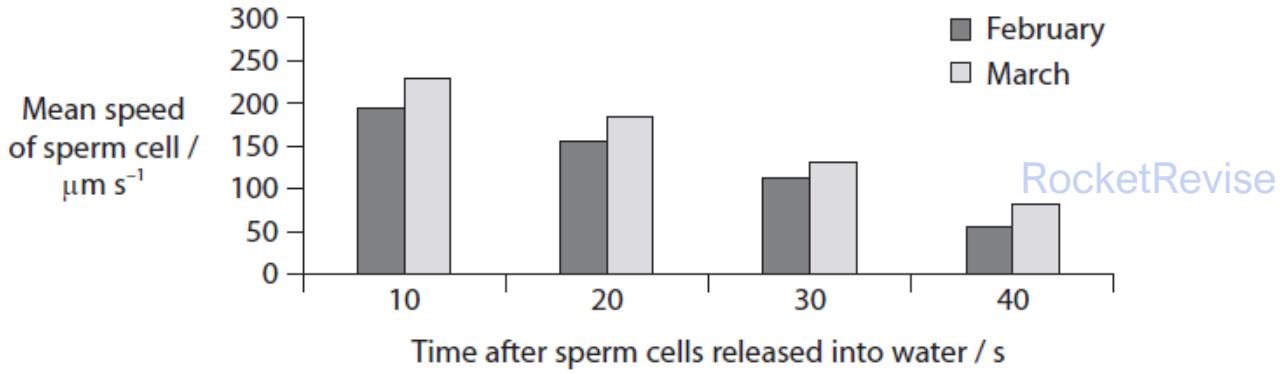
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(c) Pike sperm cells begin swimming when they are released into water.

A scientist investigated the speed of the sperm cells after they were released into water during February and March.

The graph shows the results of this investigation.



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(i) Describe the adaptations of a sperm cell that allow movement.

(2)

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(ii) Give **two** conclusions for this investigation.

(2)

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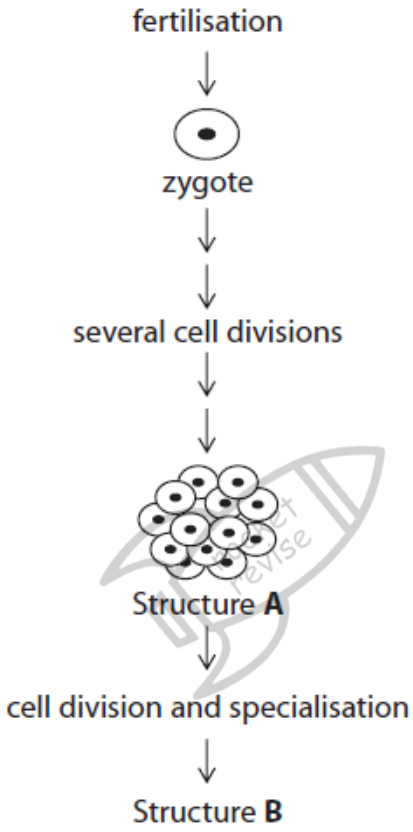
(Total for question = 12 marks)

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Q18.

The diagram shows some processes that occur after the fertilisation of an egg cell.



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(a) Structure **B** is a blastocyst.

Describe the structure of a blastocyst.

(2)

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(b) Structure **A** contained 60 cells.

The mitotic index of cells from structure **A** was calculated as 15%.

Calculate the number of cells in structure **A** which were in mitosis.

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(1)

Answer .....

(c) Structure **B** contains pluripotent stem cells.

Histone modification and DNA methylation are processes that occur when totipotent stem cells develop into pluripotent stem cells.

Explain how these two processes alter the activation of genes in the cells.

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(Total for question = 8 marks)

Q19.

Red blood cells are produced from pluripotent stem cells found in bone marrow.

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(a) Which statement about these stem cells is correct?

(1)

☒ **A** they can produce all types of cell

☒ **B** they can produce all types of cell except extraembryonic cells

☒ **C** they can produce some types of cell

☒ **D** they can produce red blood cells only

(b) Red blood cells contain haemoglobin.

A molecule of haemoglobin is made of four polypeptides. Each polypeptide has a haem group attached to it. The haem group is **not** made of amino acids.

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In most adult haemoglobin, there are two α polypeptides and two β polypeptides.

The diagram shows the structure of adult haemoglobin.



Describe the role of the rough endoplasmic reticulum in the synthesis of haemoglobin.

(3)

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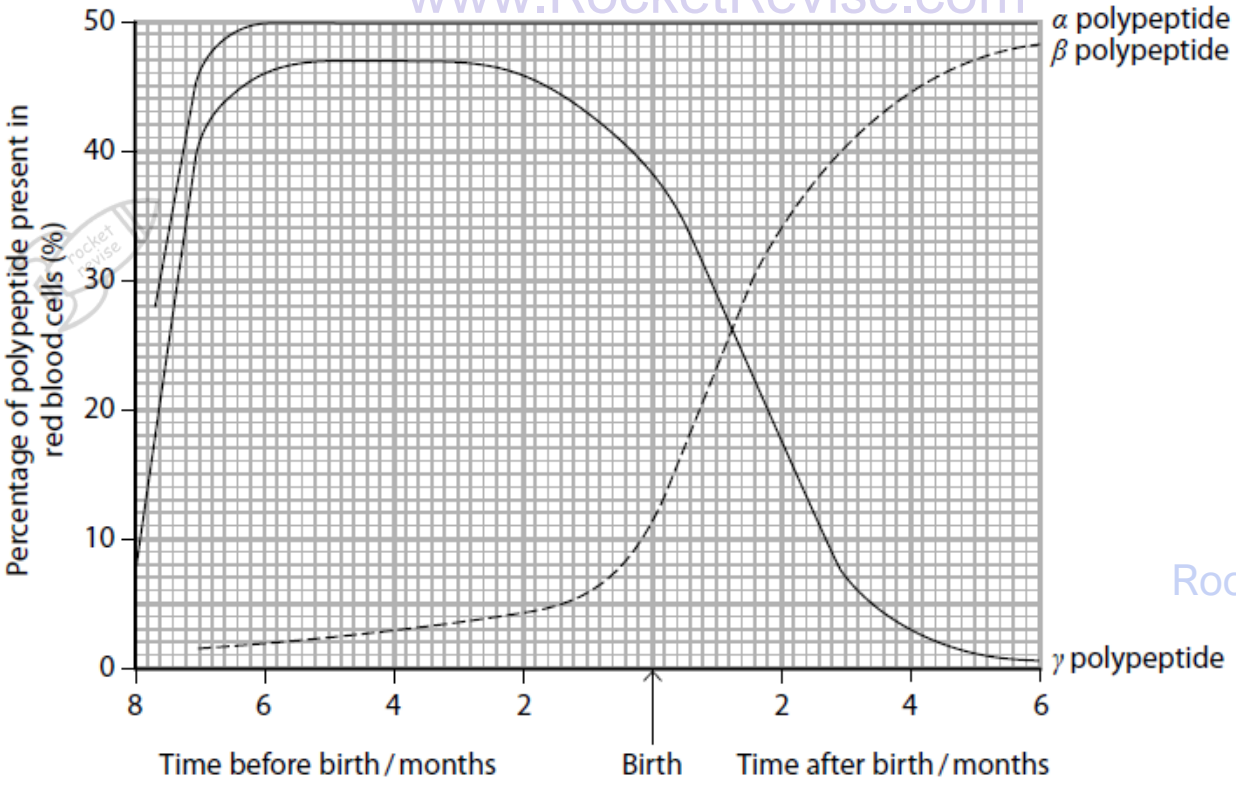
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(c) Fetal haemoglobin has a similar structure to adult haemoglobin. Fetal haemoglobin has two α polypeptides and two γ polypeptides.

The graph shows the percentage of each polypeptide present in red blood cells in an individual before and after birth.

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(i) Describe the changes in the percentages of polypeptides present in red blood cells. Use the information in the graph to support your answer.

(3)

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(ii) Explain how epigenetic modification could result in these changes.

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(Total for question = 11 marks)

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Q20.

The photograph shows a fruit fly.



© Nature Picture Library / Alamy Stock Photo

(a) The gene for eye colour in this species of fruit fly has multiple alleles.

Homozygous fruit flies have two identical alleles for eye colour.

The table gives some information about these alleles.

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Allele	Eye colour of homozygous fruit fly	Depth of colour
+	red	darkest colour
w^{ch}	cherry	
w^{b}	blood	
w^{e}	eosin	
w^{apr}	apricot	
w^{iv}	ivory	
w^{cr}	cream	
w	white	lightest colour

RocketRevise

(i) These alleles are found at the same locus.

State what is meant by the term **locus**.

(1)

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(ii) Explain how these alleles can produce fruit flies with different coloured eyes.

(2)

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(b) The development of an embryo into either a female or a male fruit fly is determined by two different proteins.

The female specific Dsx-F protein leads to the development of a female fly and the male specific Dsx-M protein results in a male fly.

Both proteins are the product of the same gene.

Explain how this gene can give rise to either a female or a male fly.

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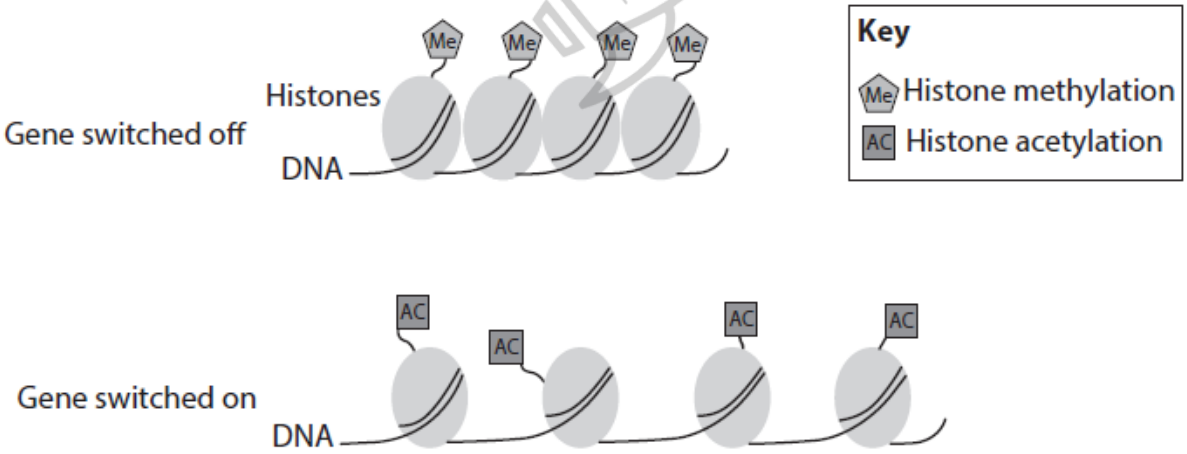
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(c) Histone modification can affect gene expression, as shown in the diagram.



Explain how histone modification can affect gene expression.

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Use the diagram to support your answer.

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(Total for question = 12 marks)

Q21.

All living organisms are made of cells.

Some cells are specialised for their function.

(a) One specialised cell is the sperm cell.

Draw a mammalian sperm cell showing the acrosome, the nucleus and a mitochondrion.

Label the acrosome, nucleus and mitochondrion.

(3)

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(b) A human egg cell is another example of a specialised cell.

(i) A human egg cell has a radius of 50 µm.

Calculate the **volume** of this human egg cell.

Use the formula:

$$V = \frac{4}{3}\pi r^3$$

(1)

Answer ..... μm^3

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(ii) The surface area of this egg cell is 31 416 μm^2.

Give the surface area to volume ratio of this egg cell.

(1)

Answer ..... :1

(iii) Explain **three** ways in which a mammalian egg cell is specialised for its functions.

(3)

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(c) A fertilised egg cell develops into a morula and then a blastocyst.

Which of the following are correct statements?

1. A blastocyst contains pluripotent stem cells.

2. A blastocyst is a hollow ball of cells whereas a morula is not.

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3. Both of these structures contain cells which divide by meiosis.

(1)

☒ **A** 1 and 2 only

☒ **B** 1 and 3 only

☒ **C** 2 and 3 only

☒ **D** 1, 2 and 3

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(Total for question = 9 marks)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In animal cells the nucleus can divide by mitosis or meiosis.

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(a) In the cell cycle, the time between two divisions of mitosis is called interphase.

(i) Describe what happens to DNA during interphase.

(2)

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(ii) Name the stage of mitosis that follows prophase.

(1)

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(b) A person with Patau syndrome had three copies of chromosome 13 in each body cell.

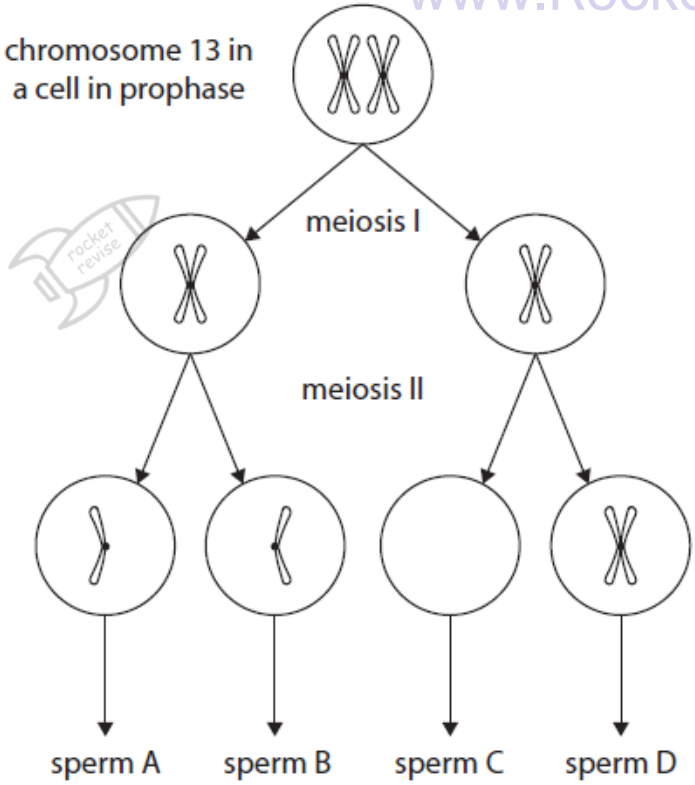
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This occurred due to an error in meiosis in the formation of his father's sperm cells.

The diagram shows what happened to chromosome 13 during the formation of the father's sperm cells.

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(i) A zygote is formed at fertilisation.

Explain how an error in meiosis could result in the production of a zygote with three copies of chromosome 13.

(4)

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(ii) The zygote would develop into a morula and then into a blastocyst.

Which row in the table is correct?

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(1)

	the zygote is:	cells in the blastocyst inner mass are:
☒ A	pluripotent	pluripotent
☒ B	pluripotent	totipotent
☒ C	totipotent	pluripotent
☒ D	totipotent	totipotent

(c) An individual can also develop Patau syndrome if part of chromosome 13 becomes attached to a different chromosome during prophase I.

Suggest a reason why part of chromosome 13 can become attached to a different chromosome when a gamete is formed.

(1)

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(d) Methylation of DNA can alter the expression of certain genes on chromosome 13.

Methylation of DNA can be passed on during mitotic cell division.

(i) Explain how DNA methylation can alter the expression of some genes.

(2)

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(ii) Explain why epigenetic changes to genes, such as DNA methylation, are passed on by mitosis.

(2)

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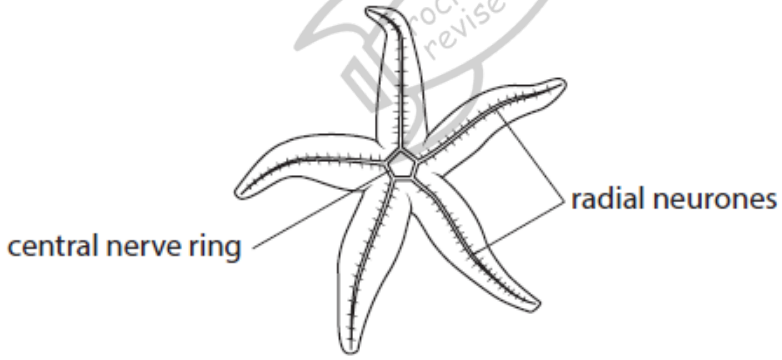
(Total for question = 13 marks)

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Q23.

Multicellular organisms, like starfish, contain many different types of cells.

(a) The diagram shows part of the nervous system of a starfish.



(Source: adapted from https://royalsocietypublishing.org/doi/pdf/10.1098/rstb.1950.0010)

(i) When nerve cells are damaged in this starfish, the gene sox2 is activated. Suggest how cell damage could lead to activation of this gene.

(2)

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(ii) Activation of the sox2 gene can stimulate body cells to become pluripotent stem cells.

Give **one** difference between a **pluripotent** stem cell and a **totipotent** stem cell.

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(1)

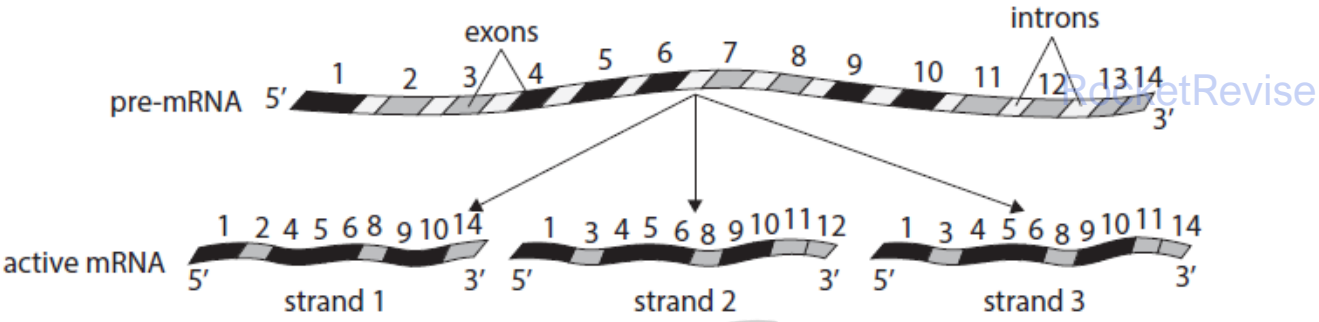
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(b) One gene from this starfish pluripotent stem cell produces pre-mRNA

The diagram shows how three different strands of active mRNA are produced from one strand of pre-mRNA.



Explain how **these** three active mRNA strands are produced.

Use the diagram to support your answer.

(4)

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(c) Society has approved the use of stem cells taken from starfish for research.

Suggest why society has approved the use of these **stem cells.**

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(Total for question = 10 marks)

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